SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA1001	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	26-10-2026	**Duration:**	60 minutes

Code of Conduct
I Jai Ganesh Rao N(192524073) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: Jai Ganesh Rao N

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:
Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Dark web threat actor de-anonymization

Introduction:

The Dark Web Threat Actor De-anonymization and Intelligence Platform is a cybersecurity solution designed to assist authorized investigators in analysing suspicious online activities. The platform uses Artificial Intelligence (AI) to analyse threat intelligence, identify potential connections between online identities, and visualize relationships between suspicious activities.

It collects available indicators such as usernames, PGP keys, cryptocurrency wallet identifiers, and infrastructure information to support investigations. The system provides relationship graphs, activity timelines, confidence scores, and intelligence reports. It aims to reduce manual investigation efforts and improve threat analysis while ensuring that findings are verified by human investigators.

Problem Description:

The identification of cybercriminals operating on the dark web is challenging because they use anonymous networks, multiple online identities, cryptocurrency wallets, and encrypted communication. Existing investigations often depend on manual analysis, which makes it difficult to identify relationships between threat actors and track suspicious activities efficiently.

The proposed Dark Web Threat Actor De-anonymization and Intelligence Platform aims to assist authorized investigators by collecting available threat intelligence, analysing relationships between online identities, identifying suspicious patterns using AI-assisted techniques, and generating intelligence reports. The system will provide relationship graphs and confidence scores to support investigations while ensuring that potential connections are verified by human analysts.

Scope of the System:

The Dark Web Threat Actor De-anonymization and Intelligence Platform is designed to help authorized cybersecurity investigators analyse threat intelligence and identify potential connections between suspicious online identities.

The system collects available threat indicators, such as usernames, PGP keys, cryptocurrency wallet identifiers, and infrastructure information. It uses AI-assisted analysis to identify patterns, visualize relationships, track suspicious activities, and generate intelligence reports.

The platform supports investigators through dashboards, relationship graphs, timelines, and confidence scores. It does not guarantee the real-world identity of threat actors; all findings must be verified by human analysts. Data collection and analysis must follow applicable laws and authorization requirements.

Functional Requirements:

The system shall provide the following functional requirements:

- User Authentication: The system shall allow authorized investigators to log in securely.
- Threat Intelligence Collection: The system shall collect available threat intelligence from authorized sources.
- Indicator Extraction: The system shall identify usernames, PGP keys, cryptocurrency wallet identifiers, and infrastructure indicators.
- Relationship Analysis: The system shall identify potential connections between online identities using shared indicators and activity patterns.
- AI-Assisted Analysis: The system shall analyse behavioural patterns and suggest possible links between threat actors.
- Relationship Visualization: The system shall display connections between online identities using relationship graphs.
- Timeline Tracking: The system shall organize suspicious activities chronologically.
- Confidence Scoring: The system shall assign confidence scores to potential connections to support investigator review.
- Dashboard: The system shall display threat indicators, suspicious relationships, and investigation summaries.

- Report Generation: The system shall generate intelligence reports and support CSV and JSON exports.
- Investigation Management: The system shall allow investigators to create, update, and review investigations.
- Audit Logging: The system shall record important user actions and maintain traceability of intelligence sources.

Non-Functional Requirements:

The system shall satisfy the following non-functional requirements:

- Performance: The system should process threat intelligence and display analysis results within a reasonable time.
- Security: The system shall protect sensitive investigation data through secure authentication and access control.
- Reliability: The system should provide consistent results and handle failures without losing stored investigation data.
- Usability: The system shall provide a simple, user-friendly interface for investigators.
- Scalability: The system should support increasing amounts of threat intelligence and additional users.
- Maintainability: The system shall be designed so that its components can be updated and maintained easily.
- Availability: The system should remain accessible to authorized users when required.
- Data Integrity: The system shall maintain the accuracy and traceability of collected intelligence.
- Privacy: The system shall restrict access to sensitive information and follow applicable privacy laws.
- Explainability: The system should provide reasons and supporting evidence for AI-generated relationship suggestions.
- Auditability: The system shall maintain audit logs of important actions and investigation activities.
- Interoperability: The system should support standard formats such as CSV and JSON for data exchange.

Actors and Use Cases:

Actors:

Actors are the users or external systems that interact with the proposed platform.

S.No.	Actor	Role
1	Administrator	Manages user accounts, permissions, and system settings.
2	Cybersecurity Investigator	Collects and analyses threat intelligence and manages investigations.
3	AI Analysis Module	Analyses indicators and suggests potential relationships between online identities.
4	Threat Intelligence Sources	Provide available threat indicators and relevant information.

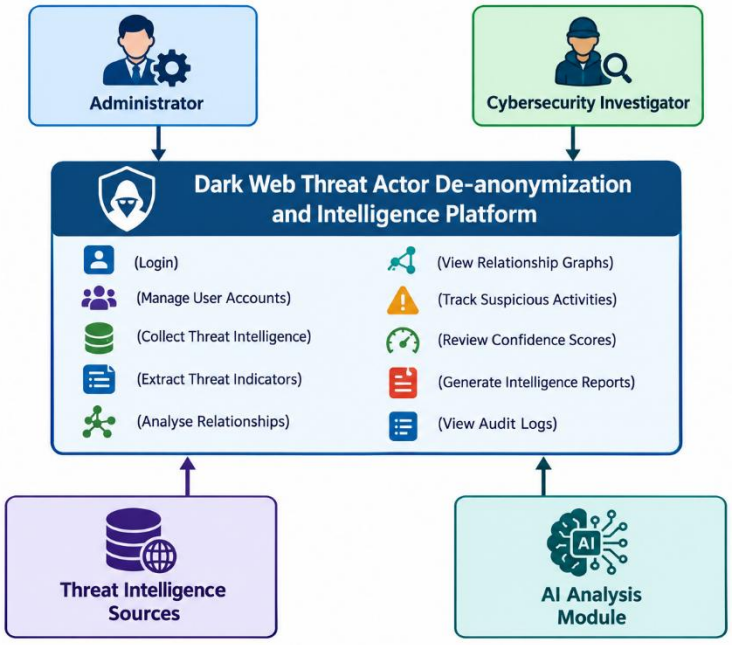
Major Use Cases:

S.No.	Use Case	Actor
1	Login to the system	Administrator, Investigator
2	Manage user accounts	Administrator
3	Collect threat intelligence	Investigator, Threat Intelligence Sources
4	Extract threat indicators	AI Analysis Module
5	Analyse relationships between identities	Investigator, AI Analysis Module
6	View relationship graphs	Investigator
7	Track suspicious activities	Investigator
8	Review confidence scores and supporting evidence	Investigator
9	Generate intelligence reports	Investigator
10	View audit logs	Administrator, Investigator

Use Case Description

The cybersecurity investigator logs in to the platform and collects threat intelligence from authorized sources. The system extracts relevant indicators and uses AI-assisted analysis to suggest potential relationships between online identities. The investigator reviews the relationship graphs, confidence scores, and supporting evidence before drawing conclusions. Finally, the investigator generates an intelligence report for further analysis.

The Use Case Diagram shows how the administrator, cybersecurity investigator, AI analysis module, and threat intelligence sources interact with the **Dark Web Threat Actor De-anonymization and Intelligence Platform**.



Explanation

- Administrator: Logs in, manages user accounts, and reviews audit logs.
- Cybersecurity Investigator: Collects intelligence, reviews potential relationships, tracks suspicious activities, and generates reports.
- AI Analysis Module: Extracts indicators and suggests potential connections between online identities.
- Threat Intelligence Sources: Supply available threat indicators to the platform.

Data Requirements:

The system shall store and manage the following data:

1. User Data: User IDs, usernames, roles, and authentication details.
2. Threat Intelligence Data: Information collected from authorized sources, including usernames, PGP keys, cryptocurrency wallet identifiers, and infrastructure indicators.
3. Relationship Data: Potential connections between online identities and the indicators supporting those connections.
4. Activity Data: Timestamps, observed activities, and event timelines.
5. Analysis Data: AI-generated suggestions, confidence scores, and supporting evidence.
6. Investigation Data: Investigation IDs, descriptions, status, and assigned investigators.
7. Report Data: Generated intelligence reports and exported files.
8. Audit Data: Records of user actions, data sources, and changes made during investigations.

The system shall protect sensitive data, maintain data integrity, and preserve source information for verification.

External Interface Requirements:

The system shall provide the following external interfaces:

1. User Interface: A web-based dashboard for investigators to manage investigations, view relationship graphs, and generate reports.
2. Software Interface: Interfaces for authorized threat intelligence sources and AI analysis components.
3. Database Interface: A secure connection to store and retrieve investigation data, threat indicators, and audit logs.
4. Data Import Interface: Support for importing permitted threat intelligence in formats such as CSV and JSON.
5. Data Export Interface: Support for exporting investigation results and reports in CSV, JSON, or PDF format.
6. Authentication Interface: A secure login interface with role-based access control.

All external connections shall follow applicable security and authorization requirements.

System Features:

The platform shall provide the following major features:

1. Secure Login: Allows authorized users to access the platform.
2. Threat Intelligence Collection: Collects available intelligence from authorized sources.
3. Indicator Extraction: Identifies usernames, PGP keys, cryptocurrency wallet identifiers, and infrastructure indicators.
4. AI-Assisted Relationship Analysis: Suggests potential links between online identities based on available evidence.
5. Relationship Graphs: Visually represents connections between indicators and online identities.
6. Timeline Analysis: Displays suspicious activities in chronological order.
7. Confidence Scoring: Provides confidence scores to help investigators prioritize potential connections.
8. Investigation Dashboard: Displays indicators, relationships, and investigation summaries.
9. Report Generation: Creates and exports intelligence reports.
10. Audit Logging: Records important actions and maintains investigation traceability.

AI-generated connections shall be treated as investigative leads and reviewed by authorized human investigators before conclusions are drawn.

Assumptions:

The development of the Dark Web Threat Actor De-anonymization and Intelligence Platform is based on the following assumptions:

1. Authorized threat intelligence sources will provide relevant and accessible data.
2. Investigators will have basic cybersecurity knowledge and system access.
3. The system will have a stable internet connection when accessing external sources.
4. AI models will assist in identifying potential relationships between online identities.
5. Collected data will be stored securely and used only for authorized investigations.
6. Investigators will verify AI-generated findings before making attribution decisions.

Constraints:

The system may face the following constraints:

1. Technical Constraints: Access to certain dark web sources may be limited, and some data may be incomplete or unavailable.
2. Security Constraints: Sensitive investigation data must be protected against unauthorized access.
3. Legal Constraints: Data collection and analysis must comply with applicable laws and authorization requirements.
4. Economic Constraints: Limited budgets may restrict access to advanced AI models and commercial threat intelligence services.
5. Operational Constraints: The accuracy of analysis depends on the quality and availability of collected data.
6. Time Constraints: The project must be developed and tested within the available academic schedule.

7. Attribution Constraints: Shared indicators do not prove that different online identities belong to the same real-world person.

Requirement Validation:

The requirements of the proposed platform will be validated to ensure that they are complete, consistent, clear, and achievable.

1. Completeness: Verify that all essential features, including intelligence collection, relationship analysis, visualization, and reporting, are included.
2. Consistency: Ensure that functional and non-functional requirements do not conflict with one another.
3. Clarity: Use clear and unambiguous statements for every requirement.
4. Feasibility: Confirm that the requirements can be implemented using the available technologies, time, and budget.
5. Security Validation: Check authentication, access control, data protection, and audit logging.
6. Accuracy Validation: Test whether the system correctly extracts indicators and identifies potential relationships using suitable test data.
7. User Validation: Obtain feedback from intended users to confirm that the dashboard and reports meet their needs.

These validation activities will help improve the quality and reliability of the system. AI-generated relationships will be treated as suggestions requiring human verification, not as confirmed identities.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25